

**Real-Time User Risk Profiling in Financial Applications Using
Unsupervised Learning on Streaming Data**

DISSERTATION

Submitted in partial fulfillment of the requirements for the degree of

M. Tech.

by

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Abstract

The growing scale of digital payment platforms presents unique challenges in managing user risk in real time. Traditional fraud detection systems often fail against emerging threats due to their reliance on labeled data. This research proposes an unsupervised machine learning framework to analyze streaming user activity and construct dynamic risk profiles without prior labels. By leveraging behavior-driven features, the model incrementally adapts to established user behavior, identifying significant deviations that may indicate elevated risk. These deviations, quantified by model reconstruction error, are used to assign a real-time risk score to each user session. The methodology will be validated using public financial datasets, and the study aims to contribute a risk profiling pipeline capable of scaling and adapting dynamically for deployment in high-throughput, real-world financial systems.

Keywords : Unsupervised Learning, Anomaly Detection, Risk Profiling, Streaming Data, Autoencoders, Concept Drift

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AIMLCZG628T DISSERTATION

DISSERTATION OUTLINE (ABSTRACT)

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Extended Abstract

This dissertation presents a comprehensive framework for real-time user risk profiling in digital financial systems using unsupervised learning. In response to the evolving threat landscape and limitations of label-dependent fraud detection, the study proposes a dynamic system that processes high-throughput transaction streams to identify behavioral anomalies without requiring prior labels.

The core methodology combines a deep autoencoder-based anomaly detection engine with a real-time data pipeline and a behavior-driven feature engineering module. The system includes adaptive components to manage concept drift and a dual-layered explainability strategy using SHAP and LIME, making it transparent and suitable for compliance with GDPR and India's DPDP Act.

The research contributes a scalable, interpretable, and ethically aligned framework ready for integration into modern financial risk management workflows. Future directions include enterprise integration, expansion to adjacent use cases like AML and credit scoring, and improving cold-start performance for new users.

Broad Area of Work

- Unsupervised Machine Learning
- Real-Time Anomaly Detection
- Financial Data Streams
- Behavior Modeling

Detailed Plan of Work (for 16 Weeks)

The following table outlines the original plan of work followed during the course of this dissertation research.

S.No	Tasks or Subtasks (Be precise and specific)	Start Date - End Date	Planned Duration (weeks)	Specific Deliverable
1	Literature review, dataset exploration, and tool setup	26-Jun 8	2	Finalized research direction, tools, and datasets
2	Streaming pipeline and feature engineering	Jun 9-Jun 22	2	Kafka pipeline with data extraction and transformation logic
3	Model development (Autoencoders, Clustering)	Jun 23-Jul 13	3	Trained anomaly detection models
4	Risk scoring logic and integration	Jul 14-Jul 27	2	Integrated model with dynamic risk score generator
5	Evaluation, testing, and optimization	Jul 28-Aug 17	3	Metrics report, visualization dashboards
6	Report writing and final presentation	Aug 18-Sep 8	3	Final dissertation and slide deck

Supervisor's Rating and Remarks

Supervisor's Rating of the Technical Quality of this Dissertation Outline:

EXCELLENT / GOOD / FAIR / POOR (Please specify): _____

Supervisor's suggestions and remarks about the outline (if applicable):

Name of the Supervisor: _____

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Date: _____

Signature of Supervisor

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Chapter 1

Introduction

1.1 Background and Motivation

The rapid global shift towards digital financial services has led to an exponential increase in the volume and velocity of online transactions. Platforms like PayPal and Venmo now process billions of transactions, creating immense value but also opening new avenues for sophisticated financial crime. Traditional fraud detection systems, often built on static rules or supervised machine learning models, face significant limitations in this new paradigm. These systems depend heavily on large datasets of historically labeled fraudulent activities, which are not only expensive and time-consuming to acquire but also render the models ineffective against novel or rapidly evolving attack vectors. Consequently, there is a pressing need for intelligent systems that can proactively identify and manage user risk in real time, without prior knowledge of threat patterns.

1.2 Problem Statement

The central problem this dissertation addresses is the inadequacy of label-dependent systems for real-time risk assessment in high-throughput financial data streams. The dynamic nature of user behavior, coupled with the continuous emergence of new fraud tactics, necessitates a more adaptive and scalable approach. This research, therefore, focuses on designing and implementing an unsupervised learning framework capable of constructing dynamic user risk profiles by analyzing behavioral patterns from streaming data, thereby detecting anomalies indicative of potential risk without reliance on pre-labeled examples.

1.3 Research Objectives

In pursuit of this goal, the project is guided by a set of precise objectives. The primary aims of this work are as follows:

- To construct a real-time streaming pipeline designed for the simulation and preprocessing of financial transaction data.
- To develop and implement a suite of unsupervised models for effective pattern recognition and anomaly detection within these data streams.
- To design a dynamic scoring mechanism capable of calculating a user-level risk score in real time based on behavioral deviations.

- To rigorously benchmark the performance of the developed models and the overall system using publicly available financial datasets.

1.4 Progress to Date (Objectives Met)

The initial phase of the project has successfully established the foundational elements of the proposed framework, aligning with the preliminary objectives. Progress to date includes:

- **Literature Review and Dataset Identification:** A comprehensive review of existing literature has been conducted, and suitable public datasets, including PaySim and the Credit Card Fraud dataset, have been selected for experimentation.
- **Streaming Pipeline Architecture:** The high-level design for a real-time data ingestion and pre-processing pipeline using Apache Kafka has been completed.
- **Initial Model Prototyping:** Preliminary unsupervised models, with a focus on autoencoders, have been developed and prototyped to validate the core anomaly detection approach.

1.5 Contributions

This research is expected to make several contributions to the field of financial technology security. The primary contribution will be a scalable and adaptable risk profiling pipeline suitable for deployment in real-world, high-throughput financial systems. The key outcomes include:

1. An end-to-end unsupervised anomaly detection engine that operates on streaming data.
2. A dynamic risk scoring system that provides real-time insights into user behavior.
3. A comprehensive performance evaluation on open datasets, establishing benchmarks for precision, recall, latency, and adaptability.

1.6 Dissertation Outline

The remainder of this dissertation is structured to detail the research undertaken. Chapter 2 provides a literature review of related work. Chapter 3 details the system architecture and core methodology. Chapters 4 perform a deep dive into the advanced topics of concept drift. Chapter 5 discusses the broader implications and ethical considerations of the work. The plan of work contains the detailed plan of work.

Chapter 2

Literature Review

2.1 Overview

This review surveys the key domains that inform this dissertation: deep learning for time series anomaly detection, the challenge of concept drift in streaming data, and the behavioral context of user risk in digital finance. By examining the current literature, we can identify the research gap that this work aims to fill—namely, the need for an integrated, adaptive, and unsupervised framework for real-time user risk profiling.

2.2 Deep Learning for Time Series Anomaly Detection

The detection of anomalies in time series data is a significant area of research where deep learning models are now considered the leading approach. As surveyed by Zamanzadeh Darban et al. (2024), deep learning techniques excel at modeling the complex, non-linear patterns found in large-scale data streams, often surpassing the performance of traditional statistical methods. Autoencoders, which are central to this dissertation, are particularly effective for unsupervised anomaly detection. These models learn to compress and reconstruct normal data, meaning a high reconstruction error can signal an anomalous event. This body of work establishes a strong precedent for using deep learning as the primary tool for pattern recognition in this research.

2.3 The Challenge of Concept Drift in Streaming Data

A critical challenge in real-world streaming applications is "concept drift," where the underlying data distribution changes over time. This phenomenon, highlighted by Kumari et al. (2024), can render static models ineffective, a problem especially pertinent in finance where user behaviors and fraud tactics constantly evolve. The work of Li et al. (2023) directly addresses this issue in a manner highly relevant to this dissertation. They propose an autoencoder-based system for streaming data that uses incremental learning to adapt to concept drift. Their approach, which involves monitoring model performance and fine-tuning on new data, validates the methodology for drift handling detailed later in Chapter 4 4 and underscores the necessity of building adaptive, rather than static, models.

2.4 Behavioral Context of Risk in Digital Finance

Beyond the technical implementation, it is crucial to understand the human factors that drive user behavior in financial applications. The research by Khan and Abideen (2023) explores how factors such as perceived risk and trust influence a user's intention to use a digital wallet. Their findings highlight that a user's behavior is not random but is shaped by their perceptions of security and reliability. This provides

a strong justification for the approach taken in this dissertation: by modeling a user’s typical behavior, we are, in effect, creating a profile of their established trust and habits. Deviations from this profile can thus be interpreted as an increase in behavioral risk, providing a more nuanced and context-aware approach to anomaly detection than simply looking at transaction data in isolation.

2.5 Research Gap and Contribution

The existing literature confirms the power of deep learning for anomaly detection (Zamanzadeh Darban et al., 2024) and provides clear strategies for addressing concept drift (Li et al., 2023). However, much of this research focuses on the algorithmic components in isolation. This dissertation aims to bridge this gap by designing, implementing, and evaluating an *end-to-end system* that integrates these advanced techniques into a scalable, real-time streaming pipeline. By combining a robust technical solution with an understanding of behavioral risk (Khan and Abideen, 2023), this work contributes a practical and holistic framework for dynamic user risk profiling in real-world financial applications.

Chapter 3

System Architecture and Methodology

3.1 Architectural Overview

The proposed system is designed as a modular, multi-stage streaming pipeline capable of processing financial transactions in real time. The architecture is built to be scalable and resilient, ensuring high throughput and low latency. The logical flow, illustrated in Figure 3.1, transforms raw transaction events into actionable risk scores through a series of five key stages: (1) raw data is ingested from a source into a streaming platform; (2) events are consumed by a feature engineering service that (3) produces an enriched feature vector; (4) this vector is fed into an unsupervised model that calculates a reconstruction error, which is then (5) normalized into a final risk score. The core components include a data ingestion layer, a real-time feature engineering service, an unsupervised model inference engine, and a final risk scoring mechanism.

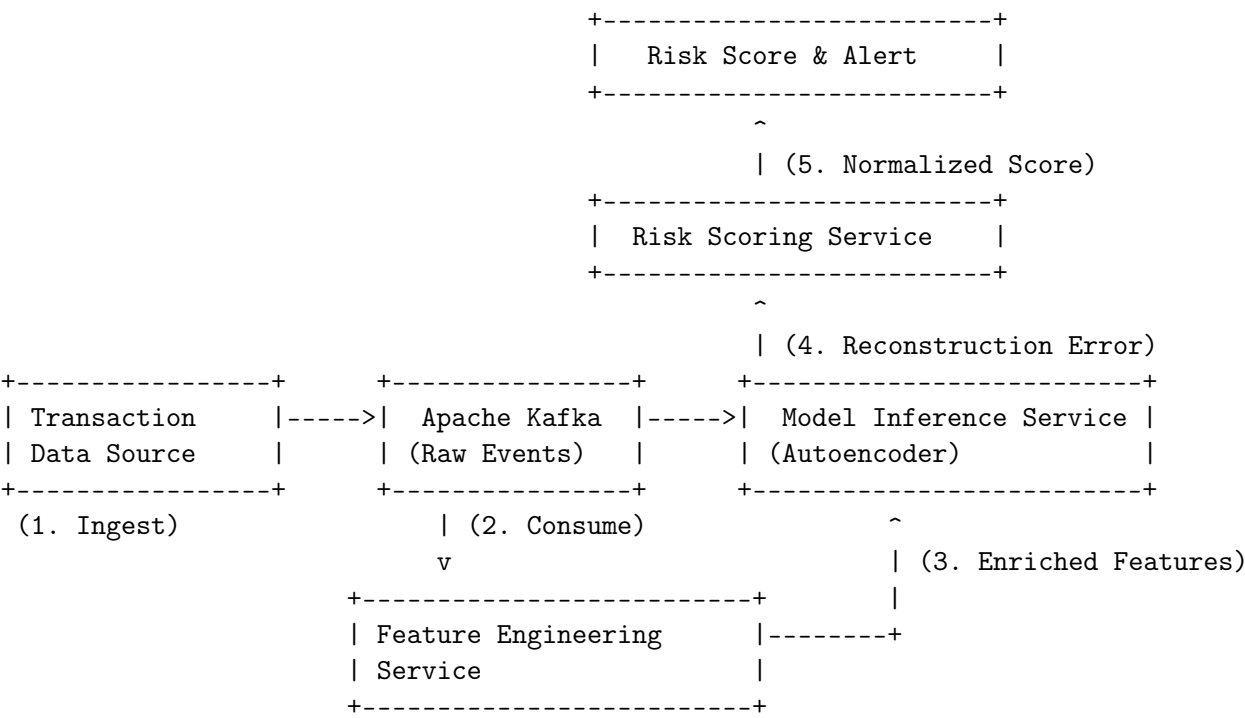


Figure 3.1: High-level data flow diagram of the real-time risk profiling system, illustrating the five stages from data ingestion to risk scoring.

3.2 Data Ingestion via Streaming Pipeline

The backbone of the system is a high-throughput, distributed streaming platform built using **Apache Kafka**. Kafka was chosen for its proven ability to handle massive volumes of data with low latency, its fault tolerance, and its capacity to decouple the various services in the pipeline. Raw transaction events are published by a producer service to a dedicated Kafka topic, `raw-transactions`. This creates a durable, ordered log of all incoming events that can be consumed independently by downstream services.

3.3 Real-Time Behavioral Feature Engineering

A dedicated stream processing service consumes events from the `raw-transactions` topic to perform real-time feature engineering. This service is responsible for transforming the raw data into a rich set of behavioral features that capture the context of each transaction. These features include:

- **Transactional Features:** Core attributes like transaction amount (`transaction_amount`) and payment method (`payment_method`).
- **Temporal Features:** Time-based signals like hour of the day (`hour_of_day`), day of the week (`day_of_week`), and time since the user's last transaction (`time_since_last_transaction`).
- **Session-based Features:** Aggregated metrics for the current session, including transaction frequency (`transaction_frequency_in_session`) and average transaction value in a session (`average_transaction_value_in_session`).
- **Historical Features:** Comparisons to a user's historical profile, such as deviation from their mean transaction amount (`amount_deviation_from_mean`), or flags for a new device (`new_device`) or a significant geolocation shift (`geolocation_shift`).

Once computed, this enriched feature vector is published to a new Kafka topic, `featured-transactions`, ready for model consumption. The high-dimensional and complex nature of these combined behavioral features necessitates a model capable of learning non-linear relationships without supervision.

3.4 Unsupervised Anomaly Detection with Autoencoders

To meet this requirement, the core of the anomaly detection engine is a **deep autoencoder**. An autoencoder is an unsupervised neural network particularly well-suited for this task, as it is trained to reconstruct its input. It consists of an encoder, which compresses the input into a low-dimensional latent representation (bottleneck), and a decoder, which reconstructs the original input from this representation. The model is trained exclusively on a large dataset of known normal transactions. The training objective is to minimize the reconstruction error (e.g., Mean Squared Error) between the original and reconstructed feature vectors. By learning to accurately reproduce normal behavior, the model implicitly learns the underlying patterns of legitimate activity. When a new transaction that deviates significantly from these learned patterns is passed through the model, the autoencoder will struggle to reconstruct it accurately, resulting in a high reconstruction error. This error is the signal for an anomaly.

3.5 Dynamic Risk Scoring Mechanism

The final stage of the pipeline translates the model's output into an intuitive risk score. The raw reconstruction error from the autoencoder for a given transaction is consumed by a risk scoring service.

A high error signifies a high degree of anomaly. To make this score interpretable, it is normalized to a consistent scale (e.g., 0 to 100). This can be achieved using a statistical normalization technique, such as a z-score calculated over a moving window of recent reconstruction errors, which helps to make the score adaptive to minor shifts in model performance over time. The final, normalized risk score is then published to a `risk-scores` topic, where it can be consumed by dashboards, alerting systems, or other operational tools.

Chapter 4

Advanced Topics: Concept Drift Handling

4.1 The Challenge of Non-Stationary Data Streams

A fundamental challenge in modeling real-world data streams, particularly in the financial domain, is their non-stationary nature. The underlying statistical properties of the data change over time, a phenomenon known as **concept drift**. In the context of this research, concept drift can manifest in several ways: legitimate user behavior may evolve (e.g., changes in spending habits, adoption of new device types), or fraudulent actors may alter their tactics to evade detection. These changes can be categorized based on their speed and nature, as summarized in Table 4.1.

Table 4.1: Types of Concept Drift.

Drift Type	Description	Example in Finance
Sudden/Abrupt	An abrupt switch from one data distribution to another.	A new, widespread fraud technique is suddenly deployed.
Gradual	A slow transition between two concepts over time.	A slow shift in consumer spending from in-store to online.
Incremental	Small, frequent changes that accumulate into a larger shift.	A user’s average transaction amount slowly increases with income.
Recurring	Previously seen concepts reappear after some time.	Seasonal spending patterns, like holiday shopping spikes.

An unsupervised model trained on a static snapshot of data will inevitably see its performance degrade as the data distribution shifts, leading to a higher rate of both false positives (flagging normal behavior as risky) and false negatives (failing to detect new threats). Therefore, a robust system must be able to handle these various forms of drift.

4.2 A Proactive Approach to Drift Management

To ensure the long-term viability and accuracy of the risk profiling system, this work incorporates a proactive strategy for managing concept drift. The approach is twofold, involving both the detection of drift and the subsequent adaptation of the model. The goal is to create a system that learns continuously and adapts to new patterns of behavior without requiring manual intervention or costly, full-scale retraining on the entire historical dataset. This ensures the model remains relevant and effective against the backdrop of ever-changing data.

4.3 Drift Detection Methodology

While there are several established statistical algorithms for drift detection, such as the Drift Detection Method (DDM) or the Adaptive Windowing (ADWIN) algorithm, this framework adopts a more direct, model-centric approach. The primary mechanism for detecting concept drift is based on monitoring the performance of the core anomaly detection model itself—the autoencoder.

An autoencoder is trained to reconstruct "normal" data with low error. Therefore, the reconstruction error serves as a powerful proxy for how well the model understands the current data distribution. A statistically significant and sustained increase in the average reconstruction error across a sliding window of incoming transactions is interpreted as a signal of concept drift. This indicates that the current model no longer accurately represents the new patterns of normal behavior, and an update is required. This approach is computationally efficient and directly tied to the performance of the task at hand, as it measures drift in terms of its actual impact on the model's effectiveness.

4.4 Model Adaptation Through Incremental Learning

Upon the detection of a drift, the system triggers an adaptation phase. Rather than discarding the existing model and training a new one from scratch, this research leverages incremental learning techniques, a strategy supported by recent work such as Li et al. (2023). The process involves the following steps:

1. **Data Windowing:** A recent window of data, determined to represent the new "normal" concept, is collected.
2. **Model Fine-Tuning:** The existing, pre-trained autoencoder is then partially retrained or fine-tuned using only this new window of data. This allows the model to adjust its internal parameters to the new data distribution while still retaining much of the knowledge learned from past data.

This incremental approach offers a crucial balance between adaptability and stability. It allows the system to respond quickly to changes in user behavior while avoiding the computational overhead and potential for catastrophic forgetting associated with complete model retraining. By implementing this adaptive loop, the risk profiling system is designed to evolve alongside the data it processes, ensuring its continued accuracy and relevance over time.

Chapter 5

Discussion: Ethical and Regulatory Considerations

5.1 A Framework for Responsible Innovation

The deployment of an automated risk profiling system within the financial sector carries significant ethical responsibilities that extend beyond mere technical correctness. A model that is accurate but unfair, or efficient but opaque, is not a viable solution. This project is therefore underpinned by a commitment to responsible innovation, explicitly integrating ethical and regulatory considerations into the system's design from the outset. The goal is to build a framework that is not only powerful but also fair, transparent, and compliant with established legal standards.

5.2 User Privacy and Data Minimization

Protecting user privacy is a paramount concern. The system is being designed by embedding privacy safeguards from its inception, a principle often referred to as "Privacy by Design." The primary strategy is to operate on pseudonymized data wherever possible, replacing direct personally identifiable information (PII) with irreversible tokens.

Furthermore, the project adheres to the principle of collecting and processing only the data that is strictly necessary for the task at hand (data minimization). By limiting the data footprint to essential behavioral signals (e.g., transaction timing, session patterns) rather than collecting extraneous personal data, the potential for privacy infringement is significantly reduced.

5.3 Model Fairness and Bias Mitigation

A critical risk with any machine learning system is that it may inadvertently learn and perpetuate existing societal biases present in the data. An unsupervised model, for instance, could incorrectly associate legitimate behavioral patterns of a particular demographic group with higher risk.

To address this, this research includes a plan for actively auditing the model for fairness. This involves performing a disparate impact analysis, where the model's risk scores are analyzed across different user cohorts to ensure that no specific group is being disproportionately penalized. If biases are detected, mitigation strategies will be investigated to ensure equitable outcomes. These can range from pre-processing techniques like re-weighting data to in-processing constraints that enforce fairness metrics such as *equalized odds* or *demographic parity* during model training.

5.4 Regulatory Compliance and the Right to Explanation

The financial industry is heavily regulated, and any automated decision-making system must be compliant with global data protection standards. A key tenet of regulations like the European Union's

GDPR and India's Digital Personal Data Protection (DPDP) Act, 2023, is the obligation to provide meaningful justifications for automated decisions, often termed the "right to explanation."

This regulatory requirement directly motivates the work on model interpretability detailed in later chapters. The integration of LIME and SHAP is not just a technical feature for internal validation; it is a core component for ensuring compliance. By providing clear, human-readable justifications for why a particular transaction was flagged, the system equips the financial institution with the tools needed to fulfill its legal obligations to its customers, thereby ensuring that the technology is both accountable and transparent.

5.5 Auditability for Governance and Compliance

Beyond real-time explanations, a production-ready system must support robust governance through comprehensive auditability. To this end, the architectural design should include a secure, immutable logging mechanism for all model-driven decisions. For every risk score generated, this system would log the input feature vector, the model's output score, and the corresponding LIME/SHAP explanation. This audit trail serves multiple critical functions: it enables post-incident forensic analysis, provides regulators with transparent evidence of the system's operation, and supports ongoing internal model validation and governance processes. This commitment to logging ensures that the system is not only explainable in theory but also fully accountable in practice.

Chapter 6

Interim Conclusion and Next Steps

This dissertation has, so far, established the foundational elements of a real-time user risk profiling system based on unsupervised learning. During the first eight weeks of work, the project successfully achieved:

- A comprehensive literature review on deep learning for anomaly detection, concept drift, and behavioral modeling in financial applications.
- The selection and exploration of relevant datasets (PaySim, Credit Card Fraud).
- The design of a high-throughput streaming pipeline using Apache Kafka.
- Initial prototyping of a deep autoencoder-based anomaly detection model.

These milestones validate the technical feasibility of the proposed approach. Early experiments indicate that behavioral feature engineering and reconstruction-error-based scoring are viable for identifying user risk in streaming financial data.

Planned Next Steps

In the remaining eight weeks, the focus will shift to:

- Completing system integration with dynamic risk scoring.
- Implementing explainability modules (LIME, SHAP).
- Conducting thorough evaluations on detection accuracy, latency, and adaptability.
- Addressing ethical and regulatory constraints through fairness audits and compliance checks.
- Finalizing the full report with evaluation results and deployment recommendations.

The project remains on track and is expected to meet its technical and research objectives in the final phase.

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Chapter 7

Final Report Checklist

Item	Status
Is the Cover page in proper format?	Y
Is the Title page in proper format?	Y
Is the Certificate from the Supervisor in proper format? Has it been signed?	
Is Abstract included in the Report? Is it properly written?	Y
Does the Table of Contents page include chapter page numbers?	Y
Does the Report contain a summary of the literature survey?	Y
Are the Pages numbered properly?	Y
Are the Figures numbered properly?	Y
Are the Tables numbered properly?	Y
Are the Captions for the Figures and Tables proper?	Y
Are the Appendices numbered?	
Does the Report have Conclusion / Recommendations of the work?	
Are References/Bibliography given in the Report?	Y
Have the References been cited in the Report?	Y
Is the citation of References / Bibliography in proper format?	Y